

# **Growth Lab Data Visualization Guide**

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## SECTION 1

# Foundations

Every GL chart is built from **two type families** and a **small ink ramp**. Resist adding a third family or new neutrals.

## Type families

**Source Serif 4** (serif) — chart title and chart source only. Title: 14px / weight 500 / #1A1714, always ends in a period.

**Inter** (sans-serif) — everything else inside the figure. Always use tabular-nums on numeric content.

## Chart typography

| Element         | Family                | Size | Weight | Color    | Notes                                      |
|-----------------|-----------------------|------|--------|----------|--------------------------------------------|
| Figure label    | Inter                 | 12px | 600    | #1A5A8E  | UPPERCASE, 0.14em tracking                 |
| Chart title     | Source Serif 4        | 14px | 500    | #1A1714  | Ends in a period — states a finding        |
| Chart subtitle  | Inter                 | 12px | 400    | #4F4A42  | Does not end in a period                   |
| Axis label      | Inter                 | 12px | 500    | #2C2823  | Y rotated $-90^\circ$ ; omit for year axis |
| Axis tick label | Inter                 | 12px | 400    | #2C2823  | tabular-nums always                        |
| Series label    | Inter                 | 12px | 600    | c-N-dark | Dark tone of series color (WCAG AA)        |
| Chart source    | Source Serif 4 italic | 12px | —      | #2C2823  | Always required, below chart               |

## Ink ramp

|                                                                                     |          |         |                                                       |
|-------------------------------------------------------------------------------------|----------|---------|-------------------------------------------------------|
|  | ink      | #1A1714 | Chart title, strong emphasis                          |
|  | ink-2    | #2C2823 | Axis line, axis labels, tick labels, annotations      |
|  | ink-3    | #4F4A42 | Subtitles, secondary captions                         |
|  | accent   | #1A5A8E | Figure labels, eyebrows, links — non-data chrome only |
|  | gridline | #D8D4CC | Major gridlines inside the plot area                  |

## Axes, ticks, and gridlines

- Axis line and ticks: 1px solid #2C2823. Ticks 4px long, pointing outward — never inward.
- Tick label sits 6px outside the axis. Axis label sits 20px from the widest tick.

- Y-axis label is always rotated  $-90^\circ$ , centered on the axis. Never horizontal.
- When the X axis shows only years, omit the axis label — the ticks already name the dimension.
- Zero baseline drawn at axis weight (1px ink-2), never at gridline weight.
- Gridlines: horizontal only by default; never both X and Y unless the chart is very dense.

## Decision rules

| #  | Rule                                                  | Detail                                                                                                                                                                                                                                                 |
|----|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | <b>Seven-color palette including muted.</b>           | Colors added in order — c-1, c-2, c-3 ... Use all six categorical colors only when absolutely necessary. More than six → re-think the chart type.                                                                                                      |
| 2  | <b>Dark tone for strokes and all text.</b>            | Scatter strokes use the dark version of the fill. Every direct label, legend entry, callout, and annotation also uses the dark tone. This includes muted series: c-muted lines/bars → c-muted-dark labels. Main tones do all the fill work. (WCAG AA.) |
| 3  | <b>Overlapping marks get 0.8 opacity.</b>             | Scatter circles, area bands, overlaid polygons: fill-opacity 0.8 and stroke-opacity 0.8. Single-layer marks (bars, treemap, choropleth) stay full opacity.                                                                                             |
| 4  | <b>Axis line and ticks are 1px #2C2823.</b>           | Ticks 4px long, outward, never inward.                                                                                                                                                                                                                 |
| 5  | <b>Gridlines are 1px #D8D4CC.</b>                     | Only where the reader needs to estimate off the axis; never both X and Y unless the chart is dense. Horizontal-bar charts put gridlines on X.                                                                                                          |
| 6  | <b>Axis labels: Inter 12px / 500 / #2C2823.</b>       | Tick labels: Inter 12px / 400 / #2C2823 with tabular-nums.                                                                                                                                                                                             |
| 7  | <b>Color encodes meaning, not decoration.</b>         | If a color is not earning its keep, remove it.                                                                                                                                                                                                         |
| 8  | <b>Implement the pop-up effect whenever possible.</b> | Paint supporting data in c-muted (#AFB5BE) and reserve a saturated hue for the 1–2 series the reader must track.                                                                                                                                       |
| 9  | <b>Sequential ramp for ordered values.</b>            | Diverging only with a meaningful midpoint (gains vs. losses, above/below baseline). Never diverging for purely positive scales.                                                                                                                        |
| 10 | <b>Chart title ends with a period.</b>                | It reads as a finding statement. The subtitle does not end in a period.                                                                                                                                                                                |
| 11 | <b>No monospace.</b>                                  | All numerals use Inter with font-variant-numeric: tabular-nums.                                                                                                                                                                                        |

## SECTION 2

# Color

### Categorical palette

Seven categorical hues (six colors + one muted), each in three tones: **light / main / dark**. Colors are assigned in order — c-1 first, then c-2, c-3 ...

| Token   | Name   | Light                                                                                      | Main                                                                                       | Dark                                                                                       | Role                            |
|---------|--------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------|
| c-1     | Blue   |  #B5D5EA  |  #2F87C8  |  #1A5A8E  | Primary / single-series default |
| c-2     | Red    |  #E89C9C  |  #CC4948  |  #8A2C2B  | Contrast / lead finding         |
| c-3     | Teal   |  #92D6BF  |  #2AA584  |  #1A6B53  | Third series                    |
| c-4     | Purple |  #B5A0CC  |  #7554A3  |  #4A3470  | Fourth series                   |
| c-5     | Orange |  #F4BC8A  |  #EA822D  |  #A8580F  | Fifth series                    |
| c-6     | Yellow |  #E6E2A8  |  #CDC86B  |  #8A8638  | Sixth series                    |
| c-muted | Grey   |  #CDD2D9 |  #AFB5BE |  #5F6773 | De-emphasis / everyone else     |

### Three tones, three jobs

| Tone  | Job                                                                                                                                                                                                        | Example           |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Main  | Fills — bars, lines, treemap tiles, scatter circles.                                                                                                                                                       | Bar fill: #2F87C8 |
| Dark  | Strokes on overlapping marks and every text element tied to the series — direct labels, end-labels, legend entries, callouts. Applies to muted series too: c-muted fills → c-muted-dark labels. (WCAG AA.) | Label: #1A5A8E    |
| Light | Backgrounds, faded states, sequential ramp tail, two/three-tone stacked areas.                                                                                                                             | Band: #B5D5EA     |

### Pop-up / mute-then-highlight

Paint supporting data in **c-muted (#AFB5BE)** and reserve a saturated hue — usually c-1, sometimes c-1 + c-2 — for the 1–2 series the reader must track.

- Scatter: muted backdrop circles at 0.8 opacity + one focus circle (c-1) painted once.

- Line: muted grey lines at 2px + focus line in c-1 at 2.4px.
- Bar / treemap: muted grey bars or tiles + focus bar/tile in c-1.
- Boxplot: muted-light fill + muted stroke for boxes; focus entity as a c-1 line.
- Labels for muted marks use c-muted-dark (#5F6773). Focus labels use c-1-dark (#1A5A8E).

## Sequential vs. diverging ramps

- **Sequential** — value runs low to high with no meaningful midpoint. Single-hue ramp, darker = higher.
- **Diverging** — data has a real reference midpoint (gains vs. losses, above vs. below baseline). Never use diverging for purely positive scales.
- Default diverging pair: c-2 red ↔ c-1 blue.

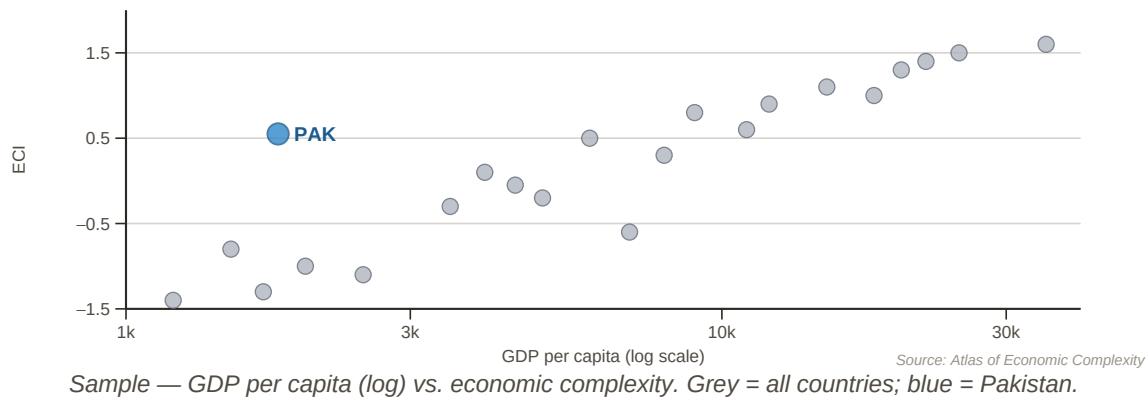
## SECTION 3

# Chart-type rules and examples

Each section below states the rules for one chart type and shows a sample using Growth Lab tokens. Apply the general rules from Sections 1–2 in all cases.

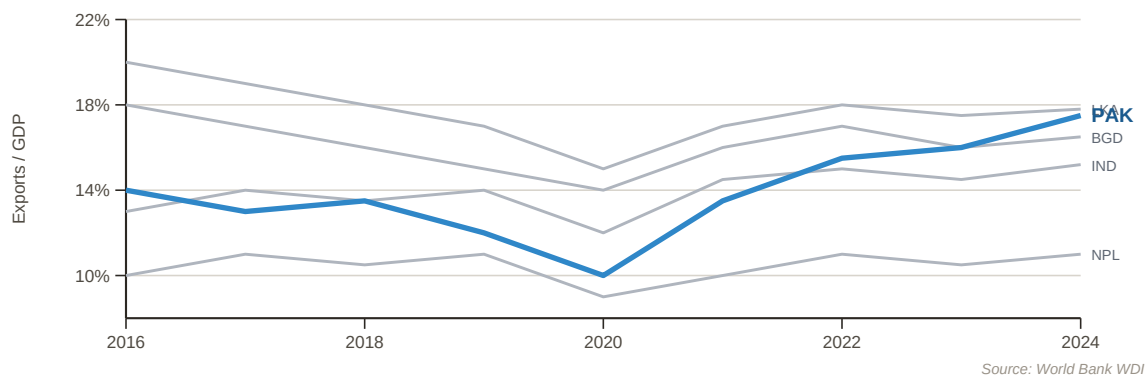
## SCATTER PLOT

- Circles only. Fill = main tone (c-N), fill-opacity 0.8. Stroke = dark tone (c-N-dark), stroke-opacity 0.8, 1px.
- Pop-up pattern: all backdrop points in c-muted / c-muted-dark at 0.8 opacity. Focus point painted once — excluded from the muted layer.
- X-axis: use log scale when encoding GDP per capita or any order-of-magnitude variable.
- Direct labels in the dark tone of the series; use repel to avoid overlap.



## LINE CHART

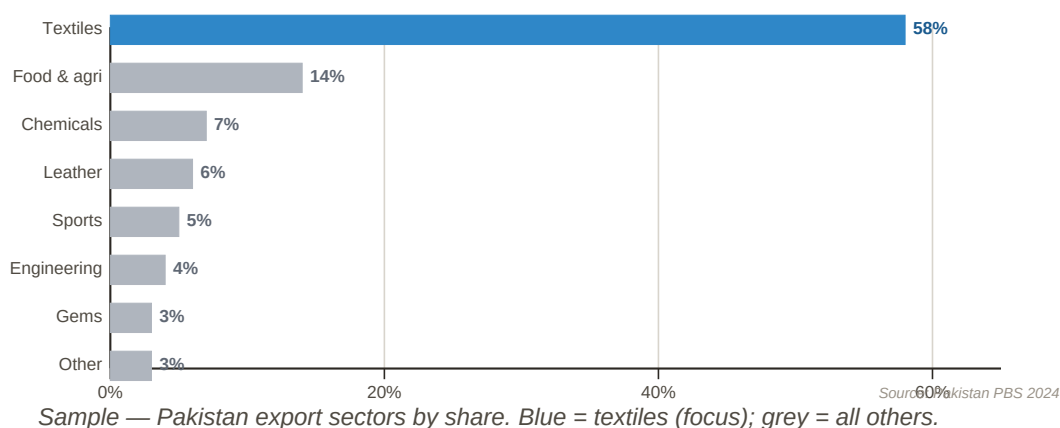
- Line weight: 2px standard; 2.4px for the highlighted focus series. stroke-linejoin: round.
- Solid lines only. Dashed lines reserved for projections or forecasts.
- Pop-up: muted lines (c-muted, 2px) for backdrop; focus line (c-1, 2.4px) on top.
- Direct end-of-line labels preferred over a legend when 1–4 series are shown. Labels use the dark tone.



Sample — exports as % of GDP, 2016–2024. Muted = peer countries; blue = Pakistan.

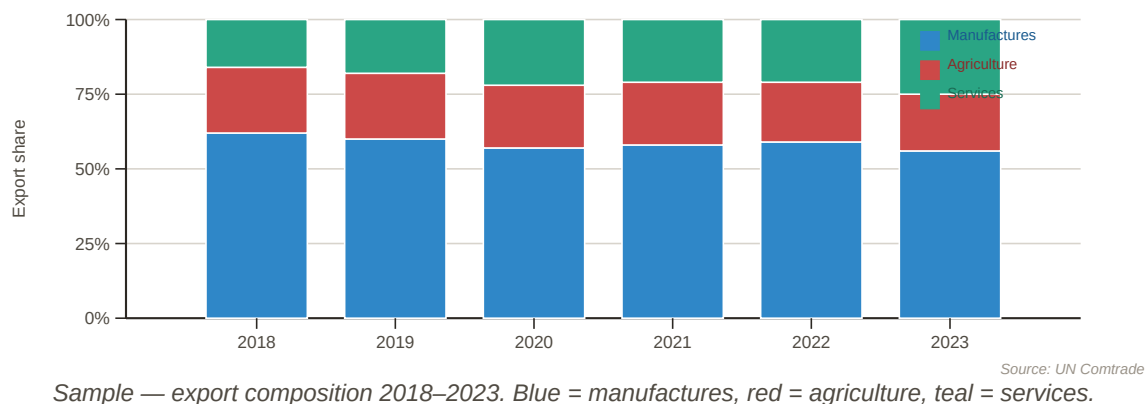
## BAR CHART (VERTICAL AND HORIZONTAL)

- Fill = main tone, full opacity, no stroke. Bars abut directly.
- Pop-up: all bars in c-muted; focus bar in c-1 (or c-2 for lead-finding contrast).
- Zero baseline drawn at axis weight (ink-2), not gridline weight.
- Value labels use the dark tone of the bar color.



## STACKED BAR AND STACKED AREA

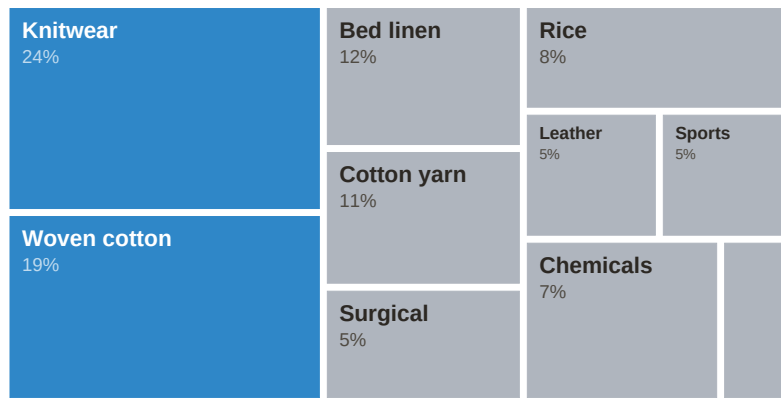
- Order categories: largest mean share at the bottom, upward.
- Stacked bars: 1px paper-white gap between each segment.
- Colors assigned in order: c-1 bottom, c-2 middle, c-3 top.
- Two-tone option: main + light of one hue when two categories share a parent concept.



## TREEMAP

- Tile fill: main tone, full opacity, no stroke. Tiles abut directly.
- Pop-up: focus tile(s) in c-1; all others in c-muted.
- Tile label: Inter weight 600. White text on dark/saturated tiles; ink-2 on light or muted-grey tiles.

- Omit labels on tiles narrower than ~44px or shorter than ~26px.

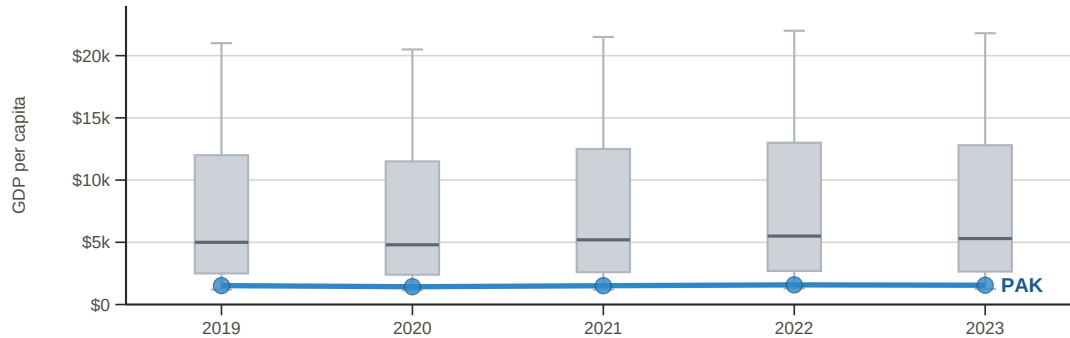


Source: Pakistan PBS / UN Comtrade 2023

Sample — Pakistan export product space. Blue = key textile products; grey = all others.

## BOXPLOT AND VIOLIN

- Background distribution boxes: c-muted-light fill + c-muted stroke, 1px. Median line: c-muted-dark, 1.5px.
- Focus entity (single country, cohort): c-1 line at 2.4px on top of the boxes.
- Focus points: fill=c-1, stroke=c-1-dark, both at 0.8 opacity, radius 5px.
- Whiskers mark the 10th–90th percentile range, with small end-caps.
- End-labels: focus entity → c-1-dark; peer median label → c-muted-dark.



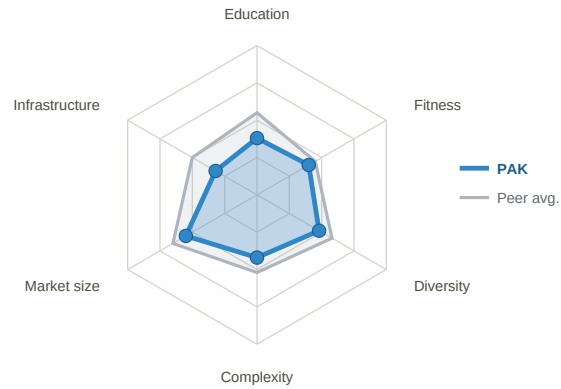
Source: World Bank WDI

Sample — regional GDP per capita distribution by year. Boxes = peer range; blue line = Pakistan.

## RADAR CHART

- Use to profile one entity across 4–8 normalized dimensions.
- Series fill: c-1 at fill-opacity 0.25 so gridlines and labels read through.
- Series stroke: c-1, full opacity, 2px, stroke-linejoin round. Vertex dots at 3px, filled c-1.
- Second comparison series: c-muted at 0.18 fill opacity, drawn under the focus series.



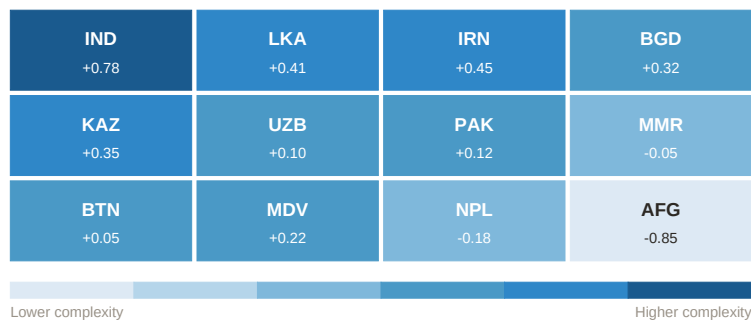


Source: WEF / World Bank

Sample — country capability profile. Blue = Pakistan; grey = peer average.

## CHOROPLETH MAP

- Sequential choropleth: single-hue ramp, darker = higher value. Full opacity; 0.5px stroke between regions.
- Diverging choropleth: meaningful midpoint only. Red tail = negative, blue = positive.
- Never use a rainbow palette — it introduces false boundaries and misleads colour-blind readers.
- Include a legend bar with labeled endpoints and midpoint. Source line required below the map.



Source: Atlas of Economic Complexity

Sample — economic complexity index, South Asia. Tile grid encodes sequential blue ramp.